



xScape Control and Data

Interface Control Document

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List of Abbreviations

Abbreviation	Description
API	Application Programming Interface
BAT	Block Allocation Table
BBT	Bad Block Table
CE	Control Electronics
CRC	Cyclic Redundancy Check
FEE	Front-End Electronics
FPGA	Field Programmable Gate Array
I2C	Inter-Integrated Circuit
ID	Identifier
LVDS	Low Voltage Differential Signalling
N/A	Not Applicable
PPS	Pulse Per Second
SPI	Serial Peripheral Interface
TDI	Time Delayed Integration

1. Introduction

1.1 Identification

Item Description: xScape Control and Data

1.2 Intended Use

This document describes the standard xScape control and data interfaces.

It is only applicable to xScape Imagers with a software major version of 12 or newer, as reported by the IMAGER INFORMATION request.

1.3 Context and Summary

The xScape range of products (also referred to as “Imagers”) support various options for control and data interfaces. All interfaces (control and data) share a unified approach to their respective protocols across the available physical interface options, as described in this document.

Each Imager provides a control interface as well as a data interface. Both these interfaces are required to fully operate the Imager. Command and Request transactions are placed onto the Imager control interface to issue commands and request telemetry. Image data is output via the data interface.

Throughout this document, parameter types are given in C/C++ notation where:

- int8 *refers to* Signed 8-bit Integer
- uint8 *refers to* Unsigned 8-bit Integer
- int16 *refers to* Signed 16-bit Integer
- uint16 *refers to* Unsigned 16-bit Integer
- uint24 *refers to* Unsigned 24-bit Integer
- uint32 *refers to* Unsigned 32-bit Integer
- uint64 *refers to* Unsigned 64-bit Integer

2. Applicable and Referenced Documents

Table 2-1 lists documents that are referenced in this document. In the event of conflict between the contents of the reference documents and this document, this document shall take precedence.

Table 2-1: Referenced Documents

Ref. #	Reference
[1]	NXP I2C-bus specification and User Manual. Rev.6 [Online]. Available: https://www.nxp.com/docs/en/user-guide/UM10204.pdf [Accessed June 17 2021]

For undated references, the latest released version of the reference document applies. For dated references, subsequent versions of the document do not apply. It is best practice to always refer to the latest released version. Unless otherwise stated, web links referenced above were last accessed at the release date of the current version of this document.

3. Control Interface Protocol

The Control Interface Protocol defines the format of transactions on the Control Interface. The protocol supports two distinct transaction types: commands and requests. Command transactions are used to issue commands to the Imager, including parameters. Request transactions are used to request telemetry or status information from the Imager.

The Imager acts as a slave on the Control Interface. The default physical interface is I²C, but SPI may also be considered.

All transactions begin when a master writes a single ID byte to the Imager, which identifies the specific command or request. The master may then continue a command transaction by writing one or more parameter bytes to the Imager. In the case of request transactions, the master will read data bytes from the Imager.

The ID byte is partitioned as follows:

Transaction Type	ID Byte Range
Command	0x00 – 0x7F
Request	0x80 – 0xFF

Since the ID byte is unique for each command or request, it is used to determine the length and the structure of the remaining transaction bytes (payload). Little-endian mode is used for parameters and data that span multiple bytes.

Command transactions may result in immediate actions on the Imager (immediate commands) while other commands may trigger the execution of a longer task. The Imager only allows one command to be executed at a time. While a command is busy, the status of its execution may be monitored by issuing a `COMMAND STATUS` request. Subsequent incoming command transactions will be ignored until the original command has completed.

Request transactions are always concluded immediately and may be carried out even if a command is busy. Any data that can be requested will always be available.

3.1 I²C Transfers

The CE implements an I²C slave interface compliant to Standard-mode and Fast-mode as defined in [1]. An I²C master interfacing to the Imager must support the clock stretching feature.

The I²C Interface operates at a 3.3 V voltage level and supports 7-bit slave addressing. The I²C signals are buffered using an NXP PCA9517ADP device. The I²C slave address of the Imager is fixed as a build option, as per the product configuration document.

When the protocol is applied to an I²C interface, the transfers are as follows:

3.1.1 Command Transactions

Command transactions are in the form of an I²C write transfer, as shown in Figure 3-1, consisting of a mandatory Command ID byte, followed by optional parameter bytes.

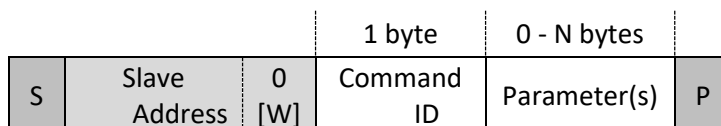


Figure 3-1: I²C Command Transaction

The fields S and P denote the I²C Start and Stop conditions, respectively. The write bit, denoted by [W], is '0'.

3.1.2 Request Transactions

Request transactions are in the form of a combined write-read transfer as shown in Figure 3-2. An I²C write transfer of the Request ID byte is immediately followed by a I²C read transfer of one or more data bytes.

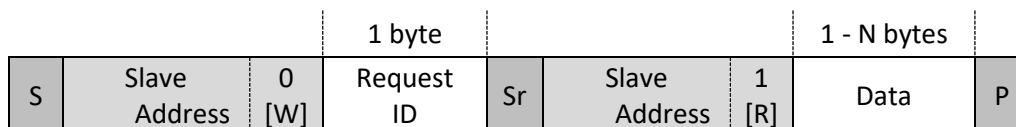


Figure 3-2: I²C Request Transaction

The fields S, Sr and P denote the I²C Start, Repeated Start, and Stop conditions, respectively. The write bit, denoted by [W], is '0' while the read bit, denoted by [R] is '1'.

3.2 SPI Transfers

The CE implements a LVCMOS (3.3V) compatible SPI slave interface consisting of four signals. The four SPI signals are:

- nSEL – Active low select signal generated by the master to select the slave interface. When not active (high) the SPI interface is not selected and placed in its reset state. All communication is initiated by the master first driving this nSEL signal low. It is kept low for the duration of the communication.
- MOSI - Master Out Slave In signal is the data output of the master.
- MISO – Master In Slave Out is the data output of the slave.
- SCK – The serial clock signal driven by the master and must have frequency between 500 kHz and 1 MHz.

The SPI interface conforms to the SPI standard “00” (clock polarity ‘0’, clock phase ‘0’). As such the values on the MOSI and MISO lines are valid at the rising edge of SCK and remain valid until the next SCK shift edge. When the protocol is applied to an SPI interface, the transfers are as follows:

3.2.1 Command Transactions

Command transactions require the SPI master to send a sequence of bytes on the MOSI line, as shown in Figure 3-3, consisting of a Command ID byte, followed by three mandatory turn-around bytes, then optional parameter bytes. The turn-around bytes may have an arbitrary value. The bytes received by the master (on the MISO line) are not used.

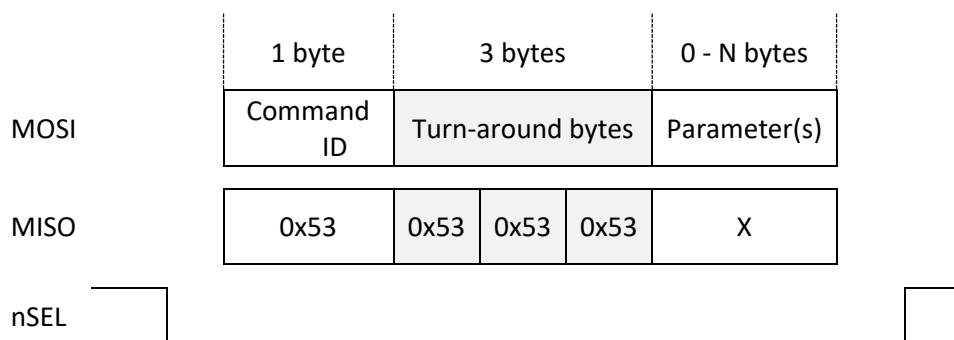


Figure 3-3: SPI Command Transaction

3.2.2 Request Transactions

Request transactions require the SPI master to send a sequence of bytes on the MOSI line, as shown in Figure 3-4, consisting of a Request ID byte, followed by three mandatory turn-around bytes, then one or more arbitrary bytes (depending on the length of the requested data). The turn-around bytes may have arbitrary value. The bytes received by the master (on the MISO line), after the turn-around bytes have been sent, represent the return data. In other words, the first 4 bytes received on the MISO line should be ignored.

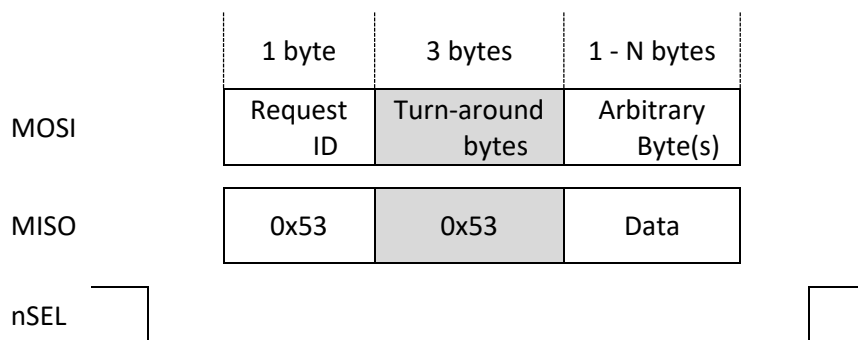


Figure 3-4: SPI Request Transaction

3.3 Command and Request Identifiers

The Command and Request Identifiers are summarized in Table 3-1 and Table 3-2. The column to the right of Table 3-1 indicates whether the respective command is an immediate command or not.

Table 3-1: Command Identifiers

Command ID	Command Name	Description	Immediate
0x00	OPEN SESSION	Generate a Session ID and open a new storage session	
0x01	ACTIVATE SESSION	Activates the session and enables storage	
0x02	CLOSE SESSION	Close the session and disables storage	
0x03	READ OUT SESSION	Initiate stored session data read out	
0x04	DELETE SESSION	Delete a session	
0x05	STORE TIME SYNC	Store Time Sync packet referenced to platform absolute time	
0x06	STORE USER DATA	Inject User data into stored image data stream	
0x07	GET SESSION INFORMATION	Retrieve information for a session	
0x08	GENERATE SESSION LIST	Generates a list of used session identifiers	Yes
0x09	GET SESSION LIST ENTRY	Retrieves a single entry from the list of sessions	
0x0A	ABORT READ OUT	Abort the current read out operation	
0x0B	DELETE ALL SESSIONS	Delete all the sessions	
0x12	READ OUT RANGE SETUP	Sets up range filtering for the next read out operation	
0x20	ENABLE SENSOR	Powers up the FEE and configures the sensor	
0x21	DISABLE SENSOR	Powers down the FEE	
0x22	SET IMAGING PARAMETER	Change an imaging parameter temporarily	Yes
0x23	SET DEFAULT IMAGING PARAMETER	Change a default imaging parameter so the new value will be loaded from the next start-up onward	
0x24	GET IMAGING PARAMETER	Retrieve an imaging parameter	Yes
0x25	RESTORE IMAGING PARAMETERS	Restore factory default imaging parameters. These will be loaded from the next start-up onward	
0x26	CONFIGURE	Configure the Imager	
0x27	CAPTURE IMAGE	Initiate image capture	
0x40	RESET	Reset the Imager	Yes
0x42	GET FEE TELEMETRY	Retrieve the FEE telemetry	
0x43	SETUP PPS	Set options for external PPS input	Yes
0x44	GET CE TELEMETRY	Retrieve the CE Telemetry	
0x45	GET OFE TELEMETRY	Retrieve the OFE Telemetry This command is not supported by all Imagers.	

Command ID	Command Name	Description	Immediate
0x46	SET DEFAULT SYSTEM PARAMETER	Change a default system parameter so the new value will be loaded from the next start-up onward	
0x47	GET SYSTEM PARAMETER	Retrieve a system parameter	Yes
0x48	RESTORE SYSTEM PARAMETERS	Restore factory default system parameters. These will be loaded from the next start-up onward	
0x49	CLEAR MONITOR COUNTERS	Clear the processor monitor counters	Yes
0x4A	SET BAD BLOCK	Mark a flash block as “bad”	
0x4B	CLEAR BAD BLOCK	Mark a flash block as “good”	
0x60	APB WRITE	Write to a firmware register (low-level)	Yes
0x61	APB READ	Read from a firmware register (low-level)	Yes
0x62	GET DATA DUMP	Retrieve binary data	

Table 3-2: Request Identifiers

Request ID	Request Name	Description
0x81	SUBSYSTEM STATES	States of Imager subsystems
0x82	COMMAND STATUS	Status of the most recent command transaction
0x83	START-UP STATUS	Status of the Imager start-up sequence
0x84	STORAGE STATUS	Status of Imager’s storage
0x85	SESSION INFORMATION	Information for the retrieved session
0x86	CURRENT SESSION ID	ID of the current open or active session
0x87	CURRENT SESSION SIZE	Size of the current open or active session
0x88	IMAGER TIME	Current Imager Time
0x89	IMAGING PARAMETER	Value of the retrieved imaging parameter
0x8A	CE TELEMETRY	Control Electronics telemetry
0x8B	FEE TELEMETRY	Front-End Electronics telemetry as retrieved
0x8C	IMAGER INFORMATION	Information relating to the Imager itself
0x8D	SESSION LIST ENTRY	Session identifier from the list of Sessions
0x8E	SYSTEM PARAMETER	Value of the retrieved system parameter
0x8F	OFE TELEMETRY	Optical Front-End telemetry as retrieved. This request is not supported by all Imagers.
0x90	RESET STATUS	Status of the last reset event
0x91	MONITOR COUNTERS	Processor Monitor related counter values
0x92	FLASH DIAGNOSTICS	Flash memory diagnostic information
0x93	SESSION DIAGNOSTICS	Session diagnostic information
various	SENSOR DIAGNOSTICS	Sensor diagnostic information
0xE0	APB DATA	APB data retrieved
0xE1	BLOCK ALLOCATION TABLE	Flash memory allocation
0xE2	BAD BLOCK TABLE	Flash memory bad blocks
0xE3	DUMP DATA	Binary data retrieved

3.4 Control Interface API

3.4.1 Session Commands

Images are captured to the onboard flash memory during a session. Each session is assigned a 32-bit Session Identifier (in the form of a rolling counter) so that the user may identify and recall specific data stored in the flash memory. A session can have one of three states:

- **Open:** A session identifier has been assigned to store data (images)
- **Active:** The flash memory has been prepared (erased) and data may be written to it
- **Closed:** The flash memory is read-only and the data within this session may be read out

The Imager can have up to 512 sessions. When this limit is reached the user must delete sessions to make room for new sessions.

3.4.1.1 Open Session

The `OPEN SESSION` command is used to generate a unique 32-bit Session Identifier for the upcoming session. This command confirms that there is room for a new Session in the directory. The Session ID may be obtained using the `CURRENT SESSION ID` request.

Open Session	
Command ID	0x00
Parameter Bytes	0
Command Parameter(s)	N/A

3.4.1.2 Activate Session

The `ACTIVATE SESSION` command permits data storage to the flash memory. This command only has an effect if the current session state is *open*. There are two ways to activate the session – automatic (imaging) mode or manual mode. This command is typically sent after the Imager has been configured and may take some time to execute, so the `SUBSYSTEM STATES` request should be queried to confirm that the command has completed.

Activate Session		
Command ID	0x01	
Parameter Bytes	4	
Command Parameter(s)	Type	Description
Manual Megabytes	uint32	Manual mode number of megabytes, or 0 for automatic mode.

The following actions are performed during execution of this command:

- A new entry is created in the Session Directory, using the Session ID.
- Flash memory preparation – a suitable number of flash blocks are located.

- Automatic mode – used for imaging sessions, requires that the Imager has been configured (to calculate the number of storage bytes required)
- Manual mode – the number of megabytes is specified manually
- A SESSION START packet is stored to the flash (refer to section 4.3.1)
- An Imager Ancillary Data packet, IMAGER INFORMATION is stored to the flash (refer to section 4.3.10.1)
- If the Imager has been configured, two more Imager Ancillary Data packets are automatically generated and stored to the flash (refer to section 4.3.10), as follows:
 - IMAGER CONFIGURATION packet (only in automatic mode)
 - SENSOR CONFIGURATION Packet (only in automatic mode)

3.4.1.3 Close Session

The CLOSE SESSION command ends the session and prevents further data from being stored to the flash memory. The Session Directory entry is updated (to specify the number of bytes stored during the session). If the session was not activated, the Session Directory is not updated, and the Session ID is not used.

Close Session	
Command ID	0x02
Parameter Bytes	0
Command Parameter(s)	N/A

3.4.1.4 Read Out Session

The READ OUT SESSION command initiates readout of the data stored during a session via the high-speed Data Interface. The command also includes packet filtering, so the output stream can be limited to only certain packet types. The packets that are read out can be enabled or disabled based on type, and the LINE DATA and THUMBNAIL LINE DATA packets can be filtered further to specific bands.

Read Out Session		
Command ID	0x03	
Parameter Bytes	10	
Command Parameter(s)	Type	Description
Session ID	uint32	The Session ID
Packet Filter	uint16	Bit mask to enable/disable readout of packet types ('1' – enabled, '0' – disabled): Bit 0 – Raw Line Data Bit 1 – Thumbnail Line Data Bit 2 – Reserved Bit 3 – User Ancillary Data Bit 4 – Imager Ancillary Data Bit 5 – Time Sync incl. PPS Time Sync Bit 6 – Video Data Bit 7 – CCSDS Stream Data Bit 8 to 15 – Reserved

Band Filter	uint32	Bit mask to include/exclude spectral band image data. The bit position is used to enable a specific band (e.g., bit 0 is for band 0). Bit n: "0" : Band n is excluded "1" : Band n is included For Snapshot Imagers, only Band 0 is applicable.
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3.4.1.5 Delete Session

The DELETE SESSION command will remove the Session Directory entry, so that the corresponding flash memory will once again be available.

Delete Session		
Command ID	0x04	
Parameter Bytes	4	
Command Parameter(s)	Type	Description
Session ID	uint32	ID of Session to delete

3.4.1.6 Store Time Sync

The STORE TIME SYNC command accepts the platform format and time as a parameter and generates a TIME SYNCHRONISATION packet which is stored as part of the current session.

Store Time Sync		
Command ID	0x05	
Parameter Bytes	9	
Command Parameter(s)	Type	Description
Platform Time Format	uint8	This field is defined in the Image Data Format section (refer to section 4.3.7).
Platform Time Value	uint64	This field is defined in the Image Data Format section (refer to section 4.3.7).

3.4.1.7 Store User Data

The STORE USER DATA command allows the platform to store ancillary data to the current session, formatted as USER ANCILLARY DATA packets (refer to section 4.3.9). The user data ID and length of the data must be provided along with the data. Since the packet payloads stored to the session must be a multiple of 32-bit, padding will automatically be applied, as well as the appending of the packet footer CRC-32.

Store User Data		
Command ID	0x06	
Parameter Bytes	varies	
Command Parameter(s)	Type	Description
User Data ID	uint8	User specified ID to identify the ancillary data
User Data Length	uint16	0 to 1024. Length of the payload data bytes to follow

User Data	varies	Maximum of 1024 bytes
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3.4.1.8 Get Session Information

The GET SESSION INFORMATION command is used to retrieve information about a closed session (i.e., not the current session). This information is subsequently made available through the SESSION INFORMATION request.

Get Session Information		
Command ID	0x07	
Parameter Bytes	4	
Command Parameter(s)	Type	Description
Session ID	uint32	ID of the session to retrieve information from

3.4.1.9 Generate Session List

The GENERATE SESSION LIST command is used to read back the list of stored sessions. The list is available via the request.

Get Session List	
Command ID	0x08
Parameter Bytes	0
Command Parameter(s)	N/A

3.4.1.10 Get Session List Entry

The GET SESSION LIST ENTRY command is used to retrieve a single session identifier from the list of sessions. The resulting value is available via the SESSION LIST ENTRY request.

Get Session List Entry	
Command ID	0x09
Parameter Bytes	0
Command Parameter(s)	N/A

3.4.1.11 Abort Read Out

The ABORT READ OUT command is used to abort a readout operation that is busy (in progress). The data output to the high-speed Data Interface will be suspended, and the Read Out State (available via the SUBSYSTEM STATES request) will change back to idle.

Abort Read Out	
Command ID	0x0A
Parameter Bytes	0
Command Parameter(s)	N/A

3.4.1.12 Delete All Sessions

The **DELETE ALL SESSIONS** command will delete all the stored sessions, resulting in the full storage capacity being available. A magic number must be passed as a parameter, which serves as a confirmation of this action which cannot be undone.

Delete all Sessions		
Command ID	0x0B	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Magic	uint8	Fixed value of 0x25

3.4.1.13 Read Out Range Setup

The **READ OUT RANGE SETUP** command sets up range filtering for the next **READ OUT SESSION** command. When enabled, the filter blocks all packets until a start condition is detected. Thereafter, packets will pass based on the standard packet filtering criteria (set in the **READ OUT SESSION** command). Packets will continue to pass until a stop condition is detected. From this point no packets will pass for the remainder of the read out. The command parameters are used to control the start and stop detection (the range). Range filtering is disabled by default and needs to be enabled explicitly by sending the **READ OUT RANGE SETUP** command before each **READ OUT SESSION** command. The range filtering pass-range is inclusive of the packets which result in the start and stop conditions.

Note that if the range start and stop bands are different, then the order in which the bands were captured should be considered.

Read Out Range Setup		
Command ID	0x12	
Parameter Bytes	17	
Command Parameter(s)	Type	Description
Mode	uint8	Range Filter Mode. 0 : Disabled (only standard filtering is applied) 1 : Filter start and stop conditions will be detected on the line number field of Line Data Packets. 2 : Filter start and stop conditions will be detected on the segment number field of CCSDS Stream Data Packets.

Range Start	uint32	Range Start Value A start condition is detected when this value appears in the specified data packet field, as well as the band field matching the specified Range Start Band. Note: A value of zero is a special case which overrides the normal start condition detection and forces the filter to pass from the beginning of the read out.
Range Stop	uint32	Range Stop Value A stop condition is detected when this value appears in the specified data packet field, as well as the band field matching the specified Range Stop Band.
Range Start Band	uint32	Range Start Band Specifies the band number for the start condition.
Range Stop Band	uint32	Range Stop Band Specifies the band number for the stop condition.

3.4.2 Imaging Commands

3.4.2.1 Enable Sensor

The `ENABLE SENSOR` command applies power to the FEE and sets up the imaging sensor.

Enable Sensor	
Command ID	0x20
Parameter Bytes	0
Command Parameter(s)	N/A



The FEE must not be powered for extended periods of time to ensure the sensor's operating temperature specification is not exceeded.

NOTICE It is recommended to enable the sensor immediately before the `CAPTURE IMAGE` command is given and disabled it directly after the capture completes.

3.4.2.2 Disable Sensor

The `DISABLE SENSOR` command powers down the FEE and imaging sensor.

Disable Sensor	
Command ID	0x21
Parameter Length	0
Command Parameter(s)	N/A

3.4.2.3 Set Imaging Parameter

The Imager has many Imaging Parameters which need to be set before an image is captured. At Imager power-up, the default imaging parameters (see 0) are automatically loaded. Thereafter, it is only necessary to set (change) those that are desired to be used with a non-default value. The SET IMAGING PARAMETER command allows a single parameter, within the list of imaging parameters, to be changed. Using this command will automatically set the 'Config State' (as reported by the SUBSYSTEM STATES request) to '0' (unconfigured).

Set Imaging Parameter		
Command ID	0x22	
Parameter Bytes	varies	
Command Parameter(s)	Type	Description
Parameter Number	uint8	Indicates the parameter to set
Parameter Value	varies	Value of the parameter

The following Imaging Parameters are available:

Parameter Number and Name		Parameter Value	
		Type	Description
General Parameters			
0x00	Thumbnail Factor	uint8	Thumbnail reduction factor 0 : Disable on-the-fly thumbnail generation 8 : 8x reduction factor in each axis 16 : 16x reduction factor in each axis 32 : 32x reduction factor in each axis 64 : 64x reduction factor in each axis The reduction factor is applied to the full sensor resolution.
0x01	Platform ID	uint16	User defined Platform ID
0x02	Instrument ID	uint16	User defined Instrument ID
0x03	Binning Factor	uint8	Binning reduction factor 0 : Disable on-the-fly binning 2 : 2x2 pixel binning 4 : 4x4 pixel binning The reduction factor is applied to the full sensor resolution.

Sensor Specific Parameters			
0x10 to 0x1f	Sensor Specific Imaging Parameters	varies	Please refer to the Imager's User Manual for the definition of these parameters.
Snapshot Parameters			
0x20	Number of Frames	uint8	1: Single frame captured 2 to m : Number of frames captured in a burst, where m is the maximum allowed as defined in the Imager's User Manual.
0x21	Frame Interval	uint32	Inter-frame time (μ s) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. The minimum value is defined in the Imager's User Manual. The maximum value is 16 seconds.
0x22	Encoding	uint8	Bits per pixel and byte/word encoding 8 : 8-bit pixels per byte 10 : 10-bit pixels (packed, 4 pixels every 5 bytes) i.e., byte sequence [0x1a, 0x2b, 0x3c, 0x4d, 0x5e] represents four 10-bit pixels [0x31a, 0x30a, 0x0d3, 0x179] 12 : 12-bit pixels (packed, 2 pixels every 3 bytes) i.e., byte sequence [0x1a, 0x2b, 0x3c] represents two 12-bit pixels [0xb1a, 0x3c2] <i>Imagers only support native 10-bit or 12-bit pixel depths for snapshots (not both).</i>
0x23	Exposure Time	uint32	Integration Time (μ s) For burst captures, exposure time must be less than Frame Interval by some margin as defined in the Imager user manual. The minimum and maximum integration time value is sensor dependent and defined in the Imager's User Manual.
0x24	Encoding Bit Offset	uint8	Bit offset used for 8-bit encoding. Selects which 8 bits of the original 10-bit pixel data are used. 0 : Upper (bits 2 to 9) 1 : Middle (bits 1 to 8) Raw pixel values > 511 will be clamped to 255. 2 : Lower (bits 0 to 7) Raw pixel values > 255 will be clamped to 255. A value of 0 is recommended
Line Scan Parameters			
0x30	Lines	uint32	Number of lines to capture for this session. The maximum value is defined in the Imager's User Manual.

0x31	Line Period	uint32	Line Period (us) (inverse of the line rate) Exposure Time (per TDI stage) is derived from or limited by this value as defined in the Imager's User Manual.
0x32	Band Setup	uint8	Spectral band number. Counts from zero (0).
		uint8	Enable band & number of TDI Stages for this band. 0 : Disable the band 1 : Enable the band for line scan without TDI 2 to m : Enable the band and perform TDI, value equals number of stages, where m is the maximum allowed as defined in the Imager's User Manual.
0x33	Band Start Row	uint8	Spectral band number. Counts from zero (0).
		uint16	Sensor Pixel Row Location on the sensor to be used for this band's effective line. This parameter is pre-programmed, and the user should not need to change this value. The value must be a multiple of 4. <i>This parameter is Read-Only for Hyperspectral Imagers for all bands, except Band 0.</i>
0x34	Scan Direction	uint8	Dependent on orbit and Imager orientation 0 : Imager scans in direction of $+x$ axis 1 : Imager scans in direction of $-x$ axis
0x35	Black Level	uint16	Black level value for TDI processing. This value is subtracted from each pixel before performing the digital TDI processing. This parameter should be fixed at the default value of 20.
0x36	Encoding	uint8	Bits per pixel and byte/word encoding 8 : 8-bit pixels per byte 12 : 12-bit pixels (packed, 2 pixels every 3 bytes) i.e., byte sequence [0x1a, 0x2b, 0x3c] represents two 12-bit pixels [0xb1a, 0x3c2] 16 : 16-bit pixels (packed, 2 pixels every 4 bytes) i.e., byte sequence [0x1a, 0x2b, 0x3c, 0x4d] represents two 12-bit pixels [0x2b1a, 0x4d3c] <i>Imagers only support 12-bit or 16-bit line scan pixel depths (not both).</i>

0x37	Encoding Bit Offset	uint8	Bit offset used for 8-bit encoding only. Selects which 8 bits of the original 12-bit or 16-bit pixel data are used. For Imager's with 12-bit line scan pixel depth: 0 : Upper (bits 4 to 11) 1 : Middle (bits 2 to 9) Raw pixel values > 1023 will be clamped to 255. 2 : Lower (bits 0 to 7) Raw pixel values > 255 will be clamped to 255. For Imager's with 16-bit line scan pixel depth: 0 : Upper (bits 8 to 15) 1 : Middle (bits 4 to 11) Raw pixel values > 4095 will be clamped to 255. 2 : Lower (bits 0 to 7) Raw pixel values > 255 will be clamped to 255. A value of 0 is recommended
0x38	Hyperspectral Band Centre Wavelength	uint8	Band number. Counts from zero (0).
		uint16	Centre Wavelength in nanometre. The BAND START ROW parameter is automatically calculated for each band based on its centre wavelength parameter. <i>This parameter is only available for Hyperspectral Imagers.</i> <i>The value read back might be different from the set value. Refer to 3.4.2.5 for details.</i>
0x39	Exposure Time	uint32	Integration Time (μs) The minimum and maximum integration time value is sensor dependent and defined in the Imager's User Manual. The maximum integration time is also limited by the LINE PERIOD parameter. A value of zero will automatically calculate and set the maximum integration time allowed for the given line period. <i>This parameter is only available for the HyperScape50 and MultiScape50 Imagers.</i>
0x3A	High-Accuracy Overhead	uint8	Overhead value for dTDI processing in high-accuracy mode. Typical value is 32. <i>This parameter is only applicable to Hyperspectral Imagers and is for Read-Only.</i>

3.4.2.4 Set Default Imaging Parameter

The default Imager parameters are stored in non-volatile memory and loaded at power-up. The

SET DEFAULT IMAGING PARAMETER command may be used to save a new default value for the applicable parameter. This will not update the current, active, configuration of the Imager since the default values are only loaded at power-up.

Set Default Imaging Parameter		
Command ID	0x23	
Parameter Bytes	varies	
Command Parameter(s)	Type	Description
Parameter Number	uint8	Indicates the parameter to set
Parameter Value	varies	Value of the parameter

3.4.2.5 Get Imaging Parameter

The GET IMAGING PARAMETER command is used to read the specified imaging parameter value. The resulting value is available via an IMAGING PARAMETER request.

Get Imaging Parameter		
Command ID	0x24	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Parameter Number	uint8	Indicates the parameter to read

All imaging parameters will return the same value as was set using either the SET IMAGING PARAMETER command or the SET DEFAULT IMAGING PARAMETER command, except the following parameter(s):

Parameter Number and Name		Parameter Value	
		Type	Description
0x38	Hyperspectral Band Centre Wavelength	uint16 x 32	List of all the band Centre Wavelengths in nanometre. The value depends on the 'Config State' as reported by the SUBSYSTEM STATES request: '0' : Imager is not configured, and the parameter value equals the centre wavelength values that was set for the bands. '1' : Imager is configured, and the parameter value equals the calculated (actual) centre wavelengths for the bands. This value may be slightly different to the set value. <i>This parameter is only available for Hyperspectral Imagers.</i>
0x39	Exposure Time	uint32	Integration Time (μs) The value depends on the 'Config State' as reported by the SUBSYSTEM STATES request: '0' : Imager is not configured, and the parameter value equals value that was set. '1' : Imager is configured, and the parameter value equals the actual integration time; value set or the automatically calculated maximum. <i>This parameter is only available for the HyperScape50 and MultiScape50 CIS Imagers.</i>

Certain imaging parameters are set per band using an index (the band number). However, when these parameters are retrieved, the full list of parameter values (for all bands) is returned:

Parameter Name	Parameter Number
Band Setup	0x32
Band Start Row	0x33
Hyperspectral Band Centre Wavelength	0x38

3.4.2.6 Restore Imaging Parameters

The RESTORE IMAGING PARAMETERS command is used to do a factory reset of all the Imaging Parameters. The non-volatile default imaging parameters are updated using the factory values. This will not update the active configuration of the Imager since the default values are only loaded at power-up.

Restore Imaging Parameters	
Command ID	0x25
Parameter Length	0
Command Parameter(s)	N/A

3.4.2.7 Configure

The `CONFIGURE` command uses the imaging parameters to configure the Imager before an image may be captured. The Imager may be configured into different modes, which determine the imaging operations, such as Snapshot or Line scan. The storage space (number of flash blocks) required is calculated (as required by the `ACTIVATE SESSION` command). If a session has been activated, and the `CONFIGURE` command is successful, two Imager Ancillary Data packets will be stored to the session (refer to section 4.3.10), as follows: `IMAGER CONFIGURATION` packet and `SENSOR CONFIGURATION` Packet. If the `CONFIGURE` command is successful, the 'Config State' (as reported by the `SUBSYSTEM STATES` request) will be set to '1' (configured).

The `CONFIGURE` command

Configure		
Command ID	0x26	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Mode	uint8	0 : Snapshot 1 : Line Scan ¹ 2 : Snapshot Test Pattern 3 : Line Scan Test Pattern ¹ 4: Video ² 5 : Line Scan High Accuracy Mode ³

1. Only supported by Line Scan Imagers (i.e., MultiScape and HyperScape). For a HyperScape Imager, this is High Band Count Mode.

2. Only supported by Snapshot Imagers (i.e., TriScape and MonoScape) that have the Video option included.

3. Only supported by HyperScape Imagers

3.4.2.8 Capture Image

The `CAPTURE IMAGE` command initiates the capture of an image for the currently active session. The Imager must be in the *configured* state and the sensor must be in the *enabled* state. The Imager capture may be triggered immediately or delayed to the next detected pulse per second (PPS). During an imaging, the Capture State may be monitored using the `SUBSYSTEM STATES` request.

Capture Image		
Command ID	0x27	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
PPS Delay	uint8	0 : immediate $n \geq 1$: Image capture will commence on n^{th} PPS pulse

3.4.3 System Commands

3.4.3.1 Reset

The `RESET` command performs a full system reset of the Imager. A magic number must be passed as a parameter, which serves as a confirmation of this action.

Reset		
Command ID	0x40	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Magic	uint8	Fixed value of 0x72

3.4.3.2 Get FEE Telemetry

The `GET FEE TELEMETRY` command is used to sample telemetry from the Front-End Electronics which houses the sensor. The sensor must be in the *enabled* state. The resulting (sampled) telemetry is available via the `FEE TELEMETRY` request.

Get FEE Telemetry	
Command ID	0x42
Parameter Length	0
Command Parameter(s)	N/A

3.4.3.3 Setup PPS

The external Pulse Per Second signal is configured through the `SETUP PPS` command. The PPS must be enabled to allow image captures with PPS delays as well as the generation of `TIME SYNCHRONISATION PPS` data packets.

Setup PPS		
Command ID	0x43	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
PPS Settings	uint8	PPS settings Bit 0 – Enable/Disable “0” : Disable “1” : Enable Bit 1 – Edge Selection “0” : Falling Edge

		"1" : Rising Edge Bit 2 – PPS Source "0" : Primary PPS Input "1" : Secondary PPS Input Bits 3 to 7 – Reserved
--	--	---

3.4.3.4 Get CE Telemetry

The GET CE TELEMETRY command is used to sample telemetry from the Control Electronics. The resulting (sampled) telemetry is available via the CE TELEMETRY request.

Get CE Telemetry	
Command ID	0x44
Parameter Length	0
Command Parameter(s)	N/A

3.4.3.5 Get OFE Telemetry

The GET OFE TELEMETRY command is used to sample telemetry from the Control Electronics. The resulting (sampled) telemetry is available via the OFE TELEMETRY request. Note that not all Imagers support this command.

Get OFE Telemetry	
Command ID	0x45
Parameter Length	0
Command Parameter(s)	N/A

3.4.3.6 Set Default System Parameter

The default system parameters are stored in non-volatile memory and loaded at power-up or after a reset. It is not possible to change any system parameter while operating the imager. The SET DEFAULT SYSTEM PARAMETER command may be used to save a new default value for the applicable parameter.

NOTICE Under normal operating conditions, it is not necessary to change the system parameters, and it is advisable to leave them at their factory defaults.

The SET DEFAULT SYSTEM PARAMETER command allows the value of a single system parameter, within the list of parameters, to be changed.

Set Default System Parameter		
Command ID	0x46	
Parameter Bytes	varies	
Command Parameter(s)	Type	Description
Parameter Number	uint8	Indicates the parameter to set
Parameter Value	varies	Value of the parameter

The following System Parameters are available:

Parameter Number and Name		Parameter Value																												
		Type	Description																											
General System Parameters																														
0x00	Flash Targets	uint16	Flash Target Enable bit mask. Each bit indicates if one of 16 targets is enabled. Under normal conditions, all bits should be set. '1' – Enabled '0' – Not enabled																											
0x01	Flash LUNs Disable	uint16	Flash LUN Disable bit mask. Each bit indicates if one of 16 LUNs is disabled. Under normal conditions, all bits should be cleared. '1' – Disabled '0' – Not disabled																											
0x02	Data Interface	uint8	High-speed Data Interface for image data packets. The hardware will not support all interfaces (this is a build option). 1 : HSDIF (Standard LVDS) 2 : <i>Reserved</i> 3 : SpaceWire																											
Protection System Parameters																														
0x20	Latch-up Channel Enable	uint8	Latch-up Channel enable bit mask. Selects which latch-up channels are enabled and may result in an over-current power-off or power-cycle of the imager. Under normal conditions, all bits should be set. <table><tr><th>Bit</th><th>Latch-up Channel</th><th>Telemetry Name</th></tr><tr><td>0</td><td>0</td><td>C_FeeSmps</td></tr><tr><td>1</td><td>1</td><td>C_FeeLdo</td></tr><tr><td>2</td><td>2</td><td>C_Brd5V0</td></tr><tr><td>3</td><td>3</td><td>C_BrdLdo</td></tr><tr><td>4</td><td>4</td><td>C_Smps3V3</td></tr><tr><td>5</td><td>5</td><td>C_Smps1V0</td></tr><tr><td>6</td><td>6</td><td>C_Smps1V2</td></tr><tr><td>7</td><td>7</td><td>C_SdramVtt</td></tr></table> '1' – Enabled '0' – Disabled	Bit	Latch-up Channel	Telemetry Name	0	0	C_FeeSmps	1	1	C_FeeLdo	2	2	C_Brd5V0	3	3	C_BrdLdo	4	4	C_Smps3V3	5	5	C_Smps1V0	6	6	C_Smps1V2	7	7	C_SdramVtt
Bit	Latch-up Channel	Telemetry Name																												
0	0	C_FeeSmps																												
1	1	C_FeeLdo																												
2	2	C_Brd5V0																												
3	3	C_BrdLdo																												
4	4	C_Smps3V3																												
5	5	C_Smps1V0																												
6	6	C_Smps1V2																												
7	7	C_SdramVtt																												
0x21	Latch-up Channel Current Limits	uint8	Channel Number: 0 to 7																											
		uint16	Current Limit in milliamperes (mA): 0 to 2047 Sets the current limit for each latch-up channel.																											

0x22	Latch-up Filter Count	uint8	Latch-up Filter oversampling count (1 to 16). Sets the number of times that over-current limit must be exceeded before latch-up is triggered (power-off or power-cycle).										
0x23	ECC Setup	uint8	Processor Monitor ECC Setup bit mask. Each bit enables a specific ECC double-bit error source, which will cause a system reset if detected. Under normal conditions, all bits should be set. <table border="1"><thead><tr><th>Bit</th><th>Description</th></tr></thead><tbody><tr><td>0</td><td>Instruction Cache</td></tr><tr><td>1</td><td>Data Cache</td></tr><tr><td>2</td><td>Bootloader RAM</td></tr><tr><td>3</td><td>Application RAM</td></tr></tbody></table> '1' – Enabled '0' – Disabled	Bit	Description	0	Instruction Cache	1	Data Cache	2	Bootloader RAM	3	Application RAM
Bit	Description												
0	Instruction Cache												
1	Data Cache												
2	Bootloader RAM												
3	Application RAM												
0x24	Watchdog Timer Enable	uint8	Watchdog Timer Enable setting. Under normal conditions, this value should be 1. 1: Enabled 0: Disabled										
0x25	Watchdog Timer Divisor	uint8	Watchdog Timer Divisor value (0 to 7). Under normal conditions, this value must remain fixed at 4.										

3.4.3.7 Get System Parameter

The GET SYSTEM PARAMETER command is used to read the specified default system parameter value. The resulting value is available via an SYSTEM PARAMETER request.

Get System Parameter		
Command ID	0x47	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Parameter Number	uint8	Indicates the parameter to read. Refer to 3.4.3.6.

3.4.3.8 Restore System Parameters

The RESTORE SYSTEM PARAMETERS command is used to do a factory reset of all the System Parameters. The non-volatile default system parameters are updated using the factory values. The updated default values will only be loaded after a power-up or reset of the Imager.

Restore System Parameters	
Command ID	0x48
Parameter Length	0
Command Parameter(s)	N/A

3.4.3.9 Clear Monitor Counters

The CLEAR MONITOR COUNTERS command is used to clear the counters associated with the processor monitor.

Restore System Parameters	
Command ID	0x49
Parameter Length	0
Command Parameter(s)	N/A

3.4.3.10 Set Bad Block

The SET BAD BLOCK command marks a specified Flash block as “bad” in the Imager’s Bad Block table. A “bad” block will not be allocated in future sessions. A magic number must be passed as a parameter, which serves as a confirmation of this action.

NOTICE Under normal operating conditions, it will not be necessary to mark flash blocks as “bad”. Only when session diagnostics identify flash block that regularly produce un-correctable errors, may this be considered.

Set Bad Block		
Command ID	0x4A	
Parameter Bytes	3	
Command Parameter(s)	Type	Description
Magic	uint8	A fixed value of 0x87.
Block Number	uint16	Indicates the block number to mark as “bad” / non-functional (0 to 2015).

3.4.3.11 Clear Bad Block

The CLEAR BAD BLOCK command marks a specified Flash block as “good” in the Imager’s Bad Block table. A “good” block will be allocated in future sessions. A magic number must be passed as a parameter, which serves as a confirmation of this action.

NOTICE Under normal operating conditions, it will not be necessary to mark flash blocks as “good”, since the Imager will be supplied with all blocks already marked “good”.

Clear Bad Block		
Command ID	0x4A	
Parameter Bytes	3	
Command Parameter(s)	Type	Description
Magic	uint8	A fixed value of 0x87.
Block Number	uint16	Indicates the block number to mark as “good” / functional (0 to 2015).

3.4.3.12 APB Write

The APB WRITE command is used to write to the low-level firmware registers of the Imager.

NOTICE Under normal operating conditions, it will not be necessary to use this command. This command should be used with caution, under the advice of the manufacturer only.

APB Write		
Command ID	0x60	
Parameter Bytes	6	
Command Parameter(s)	Type	Description
Address	uint16	Address of the register.
Data	uint32	Data to write to the register.

3.4.3.13 APB Read

The APB READ command is used to read from the low-level firmware registers of the Imager. The read data result is subsequently made available through the APB DATA request.

NOTICE Under normal operating conditions, it will not be necessary to use this command. This command should be used with caution, under the advice of the manufacturer only.

APB Read		
Command ID	0x61	
Parameter Bytes	6	
Command Parameter(s)	Type	Description
Address	uint16	Address of the register.

3.4.3.14 Get Data Dump

The GET DATA DUMP command is used to dump binary data from a specified source. The resulting data is subsequently made available through the DUMP DATA request.

NOTICE Under normal operating conditions, it will not be necessary to use this command.

Get Dump Data		
Command ID	0x62	
Parameter Bytes	2	
Command Parameter(s)	Type	Description
Source	uint8	Source of the dump data. 0: Factory System Parameters 1: Default System Parameters 2: Factory Imaging Parameters 3: Default Imaging Parameters
Length	uint8	Data length in bytes (1 to 255).

3.4.4 Requests

3.4.4.1 Subsystem States

The SUBSYSTEM STATES request returns the state of the Imager subsystems.

Subsystem States																		
Request ID	0x81																	
Data Bytes	1																	
Return Data	Type	Description																
Status	uint8	<table><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Res</td><td>Rd</td><td colspan="2">Capture</td><td>Sen</td><td>Cfg</td><td colspan="2">Session</td></tr></table>	7	6	5	4	3	2	1	0	Res	Rd	Capture		Sen	Cfg	Session	
		7	6	5	4	3	2	1	0									
		Res	Rd	Capture		Sen	Cfg	Session										

- Bits 1:0 - Session State
 - "00" : Closed
 - "01" : Open
 - "10" : Active (ready)
 - "11" : Reserved
- Bit 2 - Config State
 - "0" : Unconfigured
 - "1" : Successfully Configured
- Bit 3 - Sensor State
 - "0" : Disabled (Powered Off)
 - "1" : Enabled (Powered On)
- Bits 5:4 - Capture State
 - "00" : Idle
 - "01" : Waiting (for PPS)
 - "10" : Busy
 - "11" : Reserved
- Bit 6 - Read Out State
 - "0" : Idle

- “1” : Busy
- Bit 7 - Reserved

3.4.4.2 Command Status

The `COMMAND STATUS` request returns the status of the most recent command transaction on the Control Interface.

Command Status		
Request ID	0x82	
Data Bytes	1	
Return Data	Type	Description
Status	uint8	Status of most recent command transaction 0 : Command executed successfully 1 : Busy 2 to 255 : Error; the command was not successful

The command errors are defined as follow:

Error ID	Error Name	Description
2	UNKNOWN_ID	Command ID is unknown (not valid)
3	COMMAND_LENGTH	Command transaction length is incorrect
4	INVALID_PARAMETER	Command Parameter is invalid (out of range)
5	SESSION_IS_OPEN	Cannot open a session that is already open or active
6	SESSION_DIR_FULL	Cannot create/open a new session (Directory full)
7	SESSION_NOT_OPEN	Cannot Active a session unless it is already open
8	SESSION_NO_CONFIG	Auto Mode activation requires configured state
9	NO_FLASH_SPACE	Space required for session is not available in flash
10	SESSION_NOT_FOUND	Session not found
11	SESSION_NOT_ACTIVE	Session not active, cannot store packet
12	USER_DATA_LENGTH	User packet data payload length incorrect
13	USER_DATA_TIMEOUT	User packet timeout storing to flash
14	FLASH_ERASE	Flash Erase failed
15	FLASH_CAPTURE	Flash Capture Setup failed
16	SESSION_INVALID	Session Checksum failed
17	SESSION_FLASH_READ	Session could not be read out as flash setup failed
18	FLASH_NO_RESP	Session read out failed as flash does not respond
19	FLASH_BUSY	Session read out failed as flash is already busy
20	CAPTURE_NO_CONFIG	Cannot capture image when not configured
24	LINES_EXCEEDED	Cannot configure as line limit is exceeded.
25	NO_BANDS_ENABLED	Cannot configure without any enabled bands
26	IMG_PARM_ERROR	Cannot configure due to invalid imaging parameters
27	BURST_EXP_INVALID	Cannot configure as exposure and frame interval are invalid.
28	LINE_EXP_INVALID	Cannot configure as exposure and line period are invalid.
29	TOTAL_TDI_EXCEEDED	Cannot configure as total TDI stages limit is exceeded.

Error ID	Error Name	Description
30	SENSOR_NOT_READY	Sensor not ready after power-up
31	SENSOR_DISABLED	Sensor is disabled, cannot capture or read telemetry
32	SENSOR_BUSY_NO_ADC	Sensor is busy and the FEE ADC is not available
33	FEE_ADC	FEE ADC sequence failed.
34	SENSOR_LVDS	Sensor LVDS channels did not initialise correctly
41	MAGIC_NUM	Incorrect magic number supplied for command
42	BAD_BLOCK_INVALID	Bad block is already in the requested state
50	SPW_NO_LINK	SpaceWire Link has not been established
70	BOOT_APP_NUM	Invalid application number
71	PROG_NO_SETUP	Cannot program before programming setup is done
72	PROG_SETUP_HEADER	Programming setup header is invalid
73	PROG_NO_SPACE	Cannot program data, no more space
74	PROG_CHECK	Programmed data checksum and header do not match
75	PROG_LENGTH	Programmed data length and header do not match
76	PROG_HEADER_WRITE	Checksum verification of written header failed
77	BOOT_HEADER_INVALID	Application header to boot is invalid
78	BOOT_PROTECT_DONE	Cannot disable protection since it is already enabled
79	BOOT_NO_MANUAL	Cannot boot manually, as automatic boot has begun

3.4.4.3 Start-up Status

The START-UP STATUS request returns the status of the start-up sequence for the Imager. It indicates if the start-up sequence is still busy and if any errors were detected during each step of the sequence. The status of each step is represented by a bit, with 0 indicating a pass, and 1 indicating a fail. If all the bits are zero, Imager start-up has completed successfully.

Start-up Status																
Request ID	0x83															
Data Bytes	1															
Return Data	Type	Description														
Start-up Status	uint8															
		<table><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td colspan="3">Reserved "000"</td><td>FIn</td><td>Ses</td><td>Img</td><td>Sys</td><td>Bsy</td></tr></table>	7	6	5	4	3	2	1	0	Reserved "000"			FIn	Ses	Img
7	6	5	4	3	2	1	0									
Reserved "000"			FIn	Ses	Img	Sys	Bsy									

Start-up Status

- Bit 0 – Bsy : Busy Flag
 - "0" : Ready
 - "1" : Busy
- Bit 1 – Sys : Load System Parameters
 - "0" : Pass
 - "1" : Fail

- Bit 2 – Img : Load Default Imaging Parameters
 - “0” : Pass
 - “1” : Fail
- Bit 3 – Ses : Load Sessions
 - “0” : Pass
 - “1” : Fail
- Bit 4 – Fln : Flash Initialisation
 - “0” : Pass
 - “1” : Fail

3.4.4.4 Storage Status

The `STORAGE STATUS` request returns the number of sessions already used, as well as the percentage of storage capacity (flash memory) that is used. The Imager may store up to 512 sessions. If the `START-UP STATUS` request indicates that the Load Sessions step failed, these values may not be accurate.

Storage Status		
Request ID	0x84	
Data Bytes	3	
Return Data	Type	Description
Sessions Used	uint16	Up to 512
Storage Capacity Used	uint8	Percentage (0 to 100)

3.4.4.5 Session Information

The `SESSION INFORMATION` request returns the session information which was previously specified by the `GET SESSION INFORMATION` command.

Session Information									
Request ID	0x85								
Data Bytes	18								
Return Data	Type	Description							
Status	uint16	Upper 8 bits							
		15	14	13	12	11	10	9	8
		Reserved							
		Lower 8 bits							
		7	6	5	4	3	2	1	0
		Reserved						STO	NC
Total Bytes	uint64	Total bytes reserved for the session							
Used Bytes	uint64	Number of bytes used by the session							

Session Status

- Bit 0 – NC : Not Closed
 - “0” : No error
 - “1” : Error. Session was not explicitly closed, and session size may be incorrect.
- Bit 1 – STO : Storage Error
 - “0” : No error
 - “1” : Error. Flash ECC errors were detected during read out. A flash block may be bad.
- Bits 2 to 15 – Reserved

3.4.4.6 Current Session ID

The CURRENT SESSION ID request returns the Session ID of the currently open or active session.

Current Session ID		
Request ID	0x86	
Data Bytes	4	
Return Data	Type	Description
Session ID	uint32	Current Session ID

3.4.4.7 Current Session Size

The CURRENT SESSION SIZE request returns the total number of bytes reserved and used for the current session.

Current Session Size		
Request ID	0x87	
Data Bytes	16	
Return Data	Type	Description
Total Bytes	uint64	Total bytes reserved for the session
Used Bytes	uint64	Number of bytes used by the session

3.4.4.8 Imager Time

The IMAGER TIME request returns the Imager runtime as a 64-bit value in microseconds.

Imager Time		
Request ID	0x88	
Data Bytes	8	
Return Data	Type	Description
Time	uint64	Current value of microsecond runtime counter

3.4.4.9 Imaging Parameter

The IMAGING PARAMETER request returns the imaging parameter previously specified by the GET IMAGING PARAMETER command.

Imaging Parameter		
Request ID	0x89	
Data Bytes	varies	
Return Data	Type	Description
Imaging Parameter	varies	Parameter data as specified by the GET IMAGING PARAMETER command

3.4.4.10 CE Telemetry

The CE TELEMETRY request returns telemetry as retrieved from the Control Electronics by the GET CE TELEMETRY command.

CE Telemetry			
Request ID	0x8A		
Data Bytes	59		
Return Data	Type	Description	Unit
V_FeeSmps	uint16	FEE Supply Voltage	mV
C_FeeSmps	uint16	FEE Supply Current	mA
C_FeeLdo	uint16	FEE Supply Current	mA
V_FeeNegSmps	int16	FEE Supply Voltage	mV
C_Brd5V0	uint16	5.0 V Supply Current	mA
V_FeeOpAmp	uint16	Fee Supply Voltage	mV
V_SdramVtt	uint16	0.6 V Supply Voltage	mV
V_Brd3V3	uint16	3.3 V Supply Voltage	mV
V_Brd2V5	uint16	2.5 V Supply Voltage	mV
V_RefCal0	uint16	ADC0 Calibration Reference Voltage	mV
V_FeeLdo	uint16	Sensor Supply Voltage	mV
V_IntTst0	uint16	ADC0 Internal Channel. Not Used.	-
V_REFM0	uint16	ADC0 Internal Channel. Not Used.	-
V_REFP0	uint16	ADC0 Internal Channel. Not Used.	-
C_BrdLdo	uint16	LDO Supply Current	mA
C_Smps3V3	uint16	3.3 V Supply Current	mA
V_Smps1V2	uint16	1.2 V Supply Voltage	mV
V_Smps1V0	uint16	1.0 V Supply Voltage	mV
C_Smps1V0	uint16	1.0 V Supply Current	mA
C_Smps1V2	uint16	1.2 V Supply Current	mA
V_Brd1V8	uint16	1.8 V Supply Voltage	mV
C_SdramVtt	uint16	0.6 V Supply Current	mA
V_RefCal1	uint16	ADC1 Calibration Reference Voltage	mV
V_IntTst1	uint16	ADC1 Internal Channel. Not Used.	-
V_REFM1	uint16	ADC1 Internal Channel. Not Used.	-
V_REFP1	uint16	ADC1 Internal Channel. Not Used.	-
V_Fpga1V0	uint16	1.0 V Supply Voltage	mV
V_Fpga1V8	uint16	1.8 V Supply Voltage	mV
V_Fpga2V5	uint16	2.5 V Supply Voltage	mV

T_Fpga	int8	FPGA Temperature	°C
--------	------	------------------	----

The nominal range for each telemetry channel can be found in the Imager's User Manual.

3.4.4.11 FEE Telemetry

The FEE TELEMETRY request returns the telemetry as retrieved from the Front-End Electronics (which houses the sensor) by the GET FEE TELEMETRY command. The telemetry response is dependent on the Imager and its definition can be found in the Imager's User Manual. The nominal range for each telemetry channel can also be found in the Imager's User Manual.

FEE Telemetry		
Request ID	0x8B	
Data Bytes	29	
Return Data	Type	Description
varies	varies	The response is Imager specific, please refer to the Imager's User Manual

3.4.4.12 Imager Information

The IMAGER INFORMATION request returns the product identification number, serial number, and version information of the Imager.

Imager Information		
Request ID	0x8C	
Data Bytes	12	
Return Data	Size (bits)	Description
Imager Product ID	uint32	Manufacturer's Product Identification Number
Imager Serial Number	uint16	Manufacturer's Product Serial Number
Firmware Major Version	uint8	Firmware Major Version
Firmware Minor Version	uint8	Firmware Minor Version
Software Major Version	uint8	Software Major Version
Software Minor Version	uint8	Software Minor Version
Baseline Number	uint16	Baseline Number for Firmware and Software

3.4.4.13 Session List Entry

The SESSION LIST ENTRY request returns a single Session identifier from the list of sessions that have previously been stored in the flash memory and are available for read out. The Generate Session List and GET SESSION LIST ENTRY commands should be sent first. Thereafter, the user should send multiple GET SESSION LIST ENTRY commands and SESSION LIST ENTRY requests to obtain the full list. If a Session identifier value of 0 is returned, this indicates that the full list has been obtained.

Session List Entry		
Request ID	0x8D	
Data Bytes	4	
Return Data	Type	Description
Session ID	uint32	Session ID from the Session list.

3.4.4.14 System Parameter

The `SYSTEM PARAMETER` request returns the system parameter previously specified by the `GET SYSTEM PARAMETER` command.

System Parameter		
Request ID	0x8E	
Data Bytes	varies	
Return Data	Type	Description
System Parameter Value	varies	Parameter data from the Parameter as specified by the <code>GET SYSTEM PARAMETER</code> command.

3.4.4.15 OFE Telemetry

The `OFE TELEMETRY` request returns the telemetry from the Optical Front-End. The telemetry response is dependent on the Imager and its definition can be found in the Imager's User Manual. Note that not all Imagers support this telemetry request. The nominal range for each telemetry channel can also be found in the Imager's User Manual.

OFE Telemetry		
Request ID	0x8F	
Data Bytes	varies	
Return Data	Type	Description
varies	varies	The response is Imager specific, please refer to the Imager's User Manual

3.4.4.16 Reset Status

The `RESET STATUS` request returns the reset status of the Imager's most recent reset event. It includes the reason for the reset, as well as the active application number and the application run time. It should be noted that the application run time is not the same as the Imager Time.

Reset Status																													
Request ID	0x90																												
Data Bytes	7																												
Return Data	Size (bits)	Description																											
Reset Reason	uint8	Reason for the most recent reset. 1: Power-Up 2: Instruction Cache un-correctable (double-bit) error 4: Data Cache un-correctable (double-bit) error 8: Bootloader RAM un-correctable (double-bit) error 16: Application RAM un-correctable (double-bit) error 32: Watchdog Time Out 64: Reset Command 128: Latch-up Power Off																											
Latch-up Flags	uint8	Latch-up flags as an 8-bit value. Each bit represents a latch-up channel, and will be set to '1' if a latch-up was detected. <table border="1"> <thead> <tr> <th>Bit</th><th>Latch-up Channel</th><th>Telemetry Name</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>C_FeeSmps</td></tr> <tr> <td>1</td><td>1</td><td>C_FeeLdo</td></tr> <tr> <td>2</td><td>2</td><td>C_Brd5V0</td></tr> <tr> <td>3</td><td>3</td><td>C_BrdLdo</td></tr> <tr> <td>4</td><td>4</td><td>C_Smps3V3</td></tr> <tr> <td>5</td><td>5</td><td>C_Smps1V0</td></tr> <tr> <td>6</td><td>6</td><td>C_Sdrm1V2</td></tr> <tr> <td>7</td><td>7</td><td>C_SdramVtt</td></tr> </tbody> </table>	Bit	Latch-up Channel	Telemetry Name	0	0	C_FeeSmps	1	1	C_FeeLdo	2	2	C_Brd5V0	3	3	C_BrdLdo	4	4	C_Smps3V3	5	5	C_Smps1V0	6	6	C_Sdrm1V2	7	7	C_SdramVtt
Bit	Latch-up Channel	Telemetry Name																											
0	0	C_FeeSmps																											
1	1	C_FeeLdo																											
2	2	C_Brd5V0																											
3	3	C_BrdLdo																											
4	4	C_Smps3V3																											
5	5	C_Smps1V0																											
6	6	C_Sdrm1V2																											
7	7	C_SdramVtt																											
Application Number	uint8	Application number of the active application. 0: Application 0 (Factory) 1: Application 1 2: Application 2 187: Bootloader																											
Run Time	uint32	Application run time in milliseconds (since reset).																											

3.4.4.17 Monitor Counters

The MONITOR COUNTERS request returns the counters associated with the processor monitor. These counters are cleared at power-up or by the CLEAR MONITOR COUNTERS command.

Monitor Counters		
Request ID	0x91	
Data Bytes	17	
Return Data	Type	Description
Instruction Cache SEC	uint16	Instruction Cache single-bit errors corrected counter.

Instruction Cache DED	uint16	Instruction Cache double-bit errors detected counter.
Data Cache SEC	uint16	Data Cache single-bit errors corrected counter.
Data Cache DED	uint16	Data Cache double-bit errors detected counter.
Bootloader RAM SEC	uint16	Bootloader RAM single-bit errors corrected counter.
Bootloader RAM DED	uint16	Bootloader RAM double-bit errors detected counter.
Application RAM SEC	uint16	Application RAM single-bit errors corrected counter.
Application RAM DED	uint16	Application RAM double-bit errors detected counter.
Watchdog Timeouts	uint8	Watchdog Timeouts (resets) counter.

3.4.4.18 Flash Diagnostics

The `FLASH DIAGNOSTICS` request returns diagnostics information about the Imager's Flash memory. This information is only useful when diagnosing Flash Initialisation problems, as indicated in the `START-UP STATUS` request.

Flash Diagnostics		
Request ID	0x92	
Data Bytes	3	
Return Data	Type	Description
Flash Status	uint8	Status of the Flash Initialisation.
Flash Failed Targets	uint16	Flash targets that failed to initialise, as a 16-bit value. Each bit represents a target number from 0 to 15. If a specific target failed, it will be set to '1'.

3.4.4.19 Session Diagnostics

The `SESSION DIAGNOSTICS` request returns diagnostics information about a specific session. The `GET SESSION INFORMATION` command must be issued before this request. The information is only useful when diagnosing session errors, as indicated in the `SESSION INFORMATION` request.

Session Diagnostics		
Request ID	0x93	
Data Bytes	5	
Return Data	Type	Description
Session Error Flags	uint8	Session error flags. If an error occurred, the appropriate bit will be set high.

			Bit	Description
			0	Capture: Block Look-up Error
			1	Capture: Single-bit error (corrected)
			2	Capture: Double-bit error (uncorrected)
			3	Capture: Timeout
			4	Readout: Block Look-up Error
			5	Readout: Single-bit error (corrected)
			6	Readout: Double-bit error (uncorrected)
			7	Readout: Timeout
Start Block	uint16	Start block of the session.		
Block Length	uint16	Length of the session in blocks.		

3.4.4.20 Sensor Diagnostics

xScape Imagers provide sensor diagnostics information which is useful when the `ENABLE SENSOR` command returns an error. The requests and telemetry responses are dependent on the Imager and are defined in the Imager's User Manual.

3.4.4.21 APB Data

The `APB DATA` request returns the data previously specified by the `APB READ` command.

NOTICE Under normal operating conditions, it will not be necessary to use this command. This command should be used with caution, under the advice of the manufacturer only.

APB Data		
Request ID	0xE0	
Data Bytes	4	
Return Data	Type	Description
Data	uint32	Data read from the register.

3.4.4.22 Block Allocation Table

The `BLOCK ALLOCATION TABLE` request returns the current Block Allocation Table (BAT) of the flash memory. The BAT is returned as an array of 252 bytes, where each bit represents a block. If the bit is set ('1'), the block is allocated. If the bit is cleared ('0'), the block is not allocated.

Block Allocation Table		
Request ID	0xE1	
Data Bytes	252	
Return Data	Type	Description
BAT Bytes	uint8 x 252	Block Allocation Table, 2016 bits. Each bit represents a flash block '0' – Not allocated (free) '1' – Allocated (used)

3.4.4.23 Bad Block Table

The BAD BLOCK TABLE request returns the current Bad Block Table (BBT) of the flash memory. The BBT is returned as an array of 252 bytes, where each bit represents a block. If the bit is set ('1'), the block is bad (non-functional). If the bit is cleared ('0'), the block is good (functional).

Bad Block Table		
Request ID	0xE2	
Data Bytes	252	
Return Data	Type	Description
BBT Bytes	uint8 x 252	Bad Block Table, 2016 bits. Each bit represents a flash block '0' – Good (functional) '1' – Bad (non-functional)

3.4.4.24 Dump Data

The DUMP DATA request returns the data which was previously specified by the GET DATA DUMP command.

Dump Data		
Request ID	0xE3	
Data Bytes	varies	
Return Data	Type	Description
Data bytes	varies	Data bytes (maximum of 255)

3.5 Bootloader API

At power-up or after a system reset, the imager's processor first runs a bootloader, which then automatically boots the default application software after a period of 500 milliseconds (ms). During this initial 500 ms period, the user may suspend the automatic booting by issuing a ENTER BOOTLOADER command. The imager's protection features may also be disabled by issuing a DISABLE PROTECTION command during this initial 500 ms period.

The bootloader allows programming of two applications – application number 1 and 2. Application number 0 is the factory application, as delivered with the Imager, and cannot be re-programmed. Application 1 and 2 are available for future software updates from the manufacturer.

The bootloader supports the following standard commands:

Command ID	Command Name
0x40	RESET
0x49	CLEAR MONITOR COUNTERS
0x60	APB WRITE
0x61	APB READ

The bootloader supports the following standard requests:

Request ID	Request Name
0x82	COMMAND STATUS
0x8C	IMAGER INFORMATION
0x90	RESET STATUS
0x91	MONITOR COUNTERS
0xE0	APB DATA

3.5.1 Bootloader Commands

The commands in this section are only applicable to the bootloader.

Command ID	Command Name	Description	Immediate
0x70	ENTER BOOTLOADER	Enter and remain in the bootloader.	Yes
0x71	DISABLE PROTECTION	Temporarily disables protection	Yes
0x72	MANUAL BOOT APPLICATION	Boots a specific application	
0x73	SET DEFAULT APPLICATION	Selects the automatically booted application	Yes
0x74	PROGRAM SETUP	Setup programming of a new application	
0x75	PROGRAM DATA	Program application data	
0x76	PROGRAM DONE	Complete the programming of the new application	
0x77	GET APPLICATION HEADER	Retrieve a specific application header	

3.5.1.1 Enter Bootloader

The ENTER BOOTLOADER command must be issued within 500 ms after power-up or a system reset. The command prevents the automatic booting of the default application, and the Imager remains in the bootloader. This allows the other bootloader commands and requests to be used.

Enter Bootloader	
Command ID	0x70
Parameter Bytes	0
Command Parameter(s)	N/A

3.5.1.2 Disable Protection

The `DISABLE PROTECTION` command temporarily disables the Latch-up, ECC and Watchdog Timer protection as specified by the parameters. This overrides the default system parameters. The protection is disabled until the next power-up or reset, where the protection is enabled according to the default system parameters.

NOTICE Under normal operating conditions, the protection should not be disabled.

Enter Bootloader		
Command ID	0x71	
Parameter Bytes	3	
Command Parameter(s)	Type	Description
Latch-up Magic Number	uint8	A value of 0x94 will disable the Latch-up protection. Any other value will enable the Latch-up protection as per the default system parameters.
ECC Magic Number	uint8	A value of 0x68 will disable the double-bit ECC protection. Any other value will enable the ECC protection as per the default system parameters.
Watchdog Magic Number	uint8	A value of 0x39 will disable the Watchdog Timer protection. Any other value will enable the Watchdog Timer protection as per the default system parameters.

3.5.1.3 Manual Boot Application

The `MANUAL BOOT APPLICATION` command boots the application number specified by the parameter. This command will not affect the default application number which specifies the application that is normally booted automatically.

Manual Boot Application		
Command ID	0x72	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Application Number	uint8	Application number to boot. 0: Application 0 (Factory) 1: User Application 1 2: User Application 2

3.5.1.4 Set Default Application

The SET DEFAULT APPLICATION command sets the default application number (in non-volatile memory). The default application number indicates which application is to be booted automatically after the Imager is powered-up or reset.

Set Default Application		
Command ID	0x73	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Application Number	uint8	Default application number to boot automatically. 0: Application 0 (Factory) 1: Application 1 2: Application 2

3.5.1.5 Program Setup

The PROGRAM SETUP command sets up the programming of an application for application number 1 or 2. It requires a software image file supplied by the manufacturer. The first 32-bytes of the software image file contain the application header, which is required as a parameter for this command.

Program Setup		
Command ID	0x74	
Parameter Bytes	33	
Command Parameter(s)	Type	Description
Application Number	uint8	Application number to program. 1: Application 1 2: Application 2
Application Header	uint8 x 32	Application header bytes (32 bytes).

3.5.1.6 Program Data

The PROGRAM DATA command writes the application binary data to the Imager's non-volatile memory. A maximum of 128 bytes can be written at a time. The PROGRAM DATA command is only allowed after a PROGRAM SETUP command has been executed successfully.

Program Data		
Command ID	0x75	
Parameter Bytes	varies	
Command Parameter(s)	Type	Description
Application Data	varies	Application Data bytes (between 1 and 128 bytes).

3.5.1.7 Program Done

The PROGRAM DONE command completes the programming of an application. This command should be called after all the application data has been written using multiple PROGRAM DATA commands. The PROGRAM DONE

command verifies the application data and if successful, the application header is written to non-volatile memory.

Program Done	
Command ID	0x76
Parameter Bytes	0
Command Parameter(s)	N/A

3.5.1.8 Get Application Header

The GET APPLICATION HEADER command is used to retrieve the application header of application 0, 1 or 2. The resulting application header is subsequently made available through the APPLICATION HEADER request.

Set Default Application		
Command ID	0x77	
Parameter Bytes	1	
Command Parameter(s)	Type	Description
Application Number	uint8	Application number. 0: Application 0 (Factory) 1: Application 1 2: Application 2

3.5.2 Bootloader Requests

The following requests are only applicable to the bootloader.

Request ID	Request Name	Description
0xF0	DEFAULT APPLICATION	Default application number
0xF1	APPLICATION HEADER	Retrieved application header
0xF2	AUTOMATIC BOOT ERROR	Automatic / default boot error

3.5.2.1 Default Application

The DEFAULT APPLICATION request returns the default application number, indicating which application is automatically booted after power-up or a system reset.

Default Application		
Request ID	0xF0	
Data Bytes	1	
Return Data	Type	Description
Default Application	uint8	Default Application Number. 0: Application 0 (Factory) 1: Application 1 2: Application 2

3.5.2.2 Application Header

The APPLICATION HEADER request returns the application header previously specific by the GET APPLICATION HEADER command.

Default Application		
Request ID	0xF1	
Data Bytes	15	
Return Data	Type	Description
Status	uint8	Status of the application header 0: Invalid 1: Valid
Magic Number	uint32	Fixed value of 0x3C9A2B51
Major Version	uint8	Application major version number
Minor Version	uint8	Application minor version number
Data Length	uint32	Length of the application data
Data Checksum	uint16	Application data checksum
Header Check Bytes	uint16	Header check bytes

3.5.2.3 Automatic Boot Error

The AUTOMATIC BOOT ERROR request returns the error code associated with a failed automatic boot-up of the default application.

Default Application		
Request ID	0xF2	
Data Bytes	1	
Return Data	Type	Description
Automatic Boot Error	uint8	Error code for a failed automatic boot-up. 0: Successful 70: Default application number is invalid 77: Default application header is invalid

4. Data Interface Protocol

4.1 Standard High-Speed LVDS Transfers

The Control Electronics implements a high-speed link transmitter, referred to as the LVDS Data Interface. It consists of three output signals: a clock, a data lane and a control lane. It also includes an input signal: a receiver ready line. The clock is free running and activated as soon as the interface is enabled for data read out. The bit rate is fixed to 100 Mbps.

4 LVDS pairs are used as follows:

- HS_Clk – Clock to which HS_D0 and HS_Ctrl is synchronised.
- HS_D0 – The data Lane.
- HS_Ctrl – Conveys synchronisation and other out-of-band status information for the entire link
- HS_RR – Optionally returned from the data sink to provide flow control

When using the standard high-speed link, the data interface has three possible states:

- OFF – The interface is turned off and all output signals (all except HS_RR) are driven low.
- INACTIVE – HS_Clk is running. HS_Ctrl has the “SYNC” bit set (‘1’) properly, however, the “LA” bit is cleared (‘0’), and no data is sent.
- ACTIVE – HS_Clk is running. HS_Ctrl has both “SYNC” and “LA” bits set (‘1’). Valid data may be sent in this state, as indicated by the “DV” bit.

Data is transferred in bytes of 8 bits, most significant bit (MSB) first, as shown in Figure 4-1.

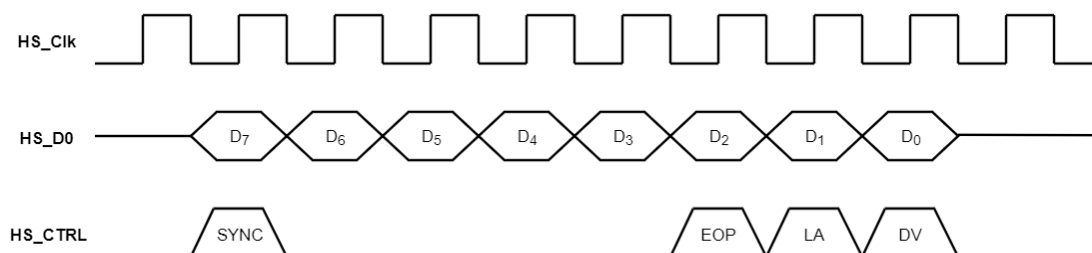


Figure 4-1: Standard High-Speed Link Byte Transfer Cycle Waveform

The information in the HS_Ctrl channel is applicable to all data lanes. The bits are defined as follow:

- SYNC – This is the byte synchronisation bit which is used to mark the start of a new byte. It is always set ('1') when the link is in any state other than the OFF state.
- EOP – The end of packet bit is set during the transfer of the last byte of a data packet. Data is formatted into packets (see Section 4.2) and this bit can be used to indicate the boundary between packets. The EOP bit will only be set when the link is in ACTIVE state and a valid byte is output (DV bit is set, '1').
- LA – This bit indicates the links state. It is set when the link is in ACTIVE state.
- DV – The data byte(s) on the data lane(s) are valid only when this bit is set ('1').

HS_RR allows the data sink to throttle the flow of data. When HS_RR is asserted ('1'), the receiver is ready to accept more data. When HS_RR is deasserted ('0') the interface is requested to stop transmitting and will do so by setting the DV bit low ('0'). The interface will wait in this state indefinitely until HS_RR is asserted ('1'). The interface may transmit up to 4 bytes once the data sink sets HS_RR low ('0'). The receiver should therefore be willing to receive up to 4 more valid bytes after setting HS_RR low ('0'). This is shown in Figure 4-2.

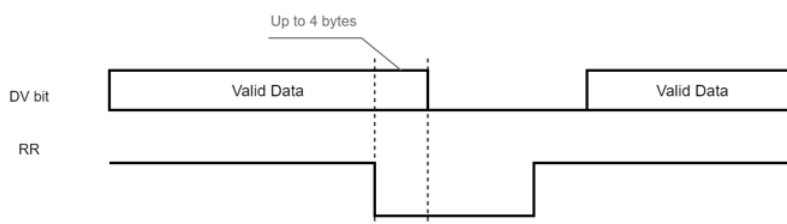


Figure 4-2: High-Speed Link HS_RR Operation

4.2 Data Format

The Image data for a session is presented in a packet-based format which is read out via the Data Interface as a continuous stream. During an imaging session, the pixel data from each sensor line is formatted into individual packets. Sensor pixel data is always packetized per line for both snapshot and line scan Imagers. Additional (ancillary) packets are also injected into the stream, in real time, so that relevant ancillary data from the user and imager may be included. User ancillary packets contain data received from the satellite bus, which typically includes attitude data, ephemeris data and time information. Imager ancillary packets allow the exact imager settings applied during the session to be stored and are automatically injected by the imager itself.

Each packet includes a header, payload and footer, as shown in Figure 4-3. The header is used to identify the packet and extract the variable length payload. The payload length must be a multiple of 4 bytes. The payload

content is uniquely defined for each identifier. Little-endian mode is used for fields that span multiple bytes. The footer is in the form of a 32-bit cyclic redundancy check (CRC-32) applied to the full packet (header and payload).

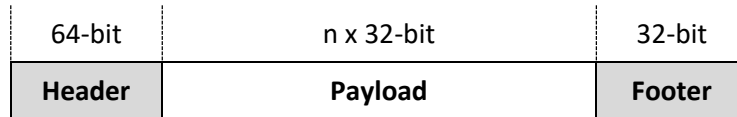


Figure 4-3: Packet Format

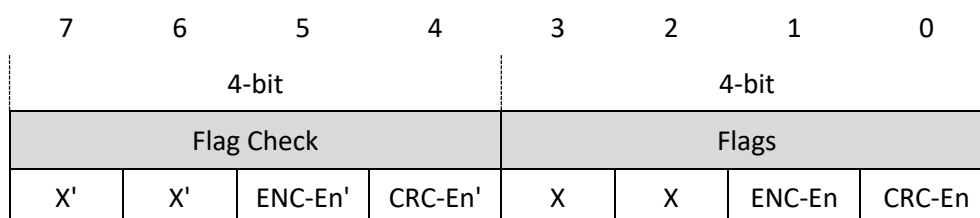
4.2.1 Header

The header size is fixed at 64 bits (8 bytes) and is formatted as shown in Table 4-1.

Table 4-1: Header Format

Field Name	Type	Description
Sync	uint16	Fixed synchronisation word ("SS" / 0x5353)
Flags	uint8	Flags to identify packet properties
Identifier	uint8	Unique identifier for each packet type
Length	uint24	Length of the payload, in bytes
Sub-packet	uint8	Non-zero when Payload is cut into smaller sub-packets

- Each packet starts with a synchronisation word, 0x5353, to assist in locating the start of a packet.
- The Flags field is defined as follows:



- The lower 4 bits represent four single bit flags
- The upper 4 bits are the complement of the lower 4 bits, which is used for error checking.
- CRC-En:
 - ♦ '1' indicates a CRC-32 footer is appended to the packet
 - ♦ '0' indicates that no footer is appended to the packet.
- ENC-En:
 - ♦ '1' indicates that the packet payload is encrypted
 - ♦ '0' indicates that the packet payload is not encrypted

- The 8-bit identifier (ID) is used to identify the packet type.
- The length field specifies the number of bytes in the payload. This must always be a multiple of 4. This means that the payload may need to be padded, and the length will include the padding bytes.
- The Sub-packet field is non-zero when the size of the payload is limited, and sub-packets are transmitted. The field is defined as follows:

2-bit		6-bit					
Sub-packet flags		Sub-packet count					
7	6	5	4	3	2	1	0
End	Start	0 to 63					

- Packets that are not sub-divided into sub-packets will have the full sub-packet field set to zero
- The first sub-packet will have start flag = '1' and count = 0
- Subsequent sub-packets will have both start and end flags = '0', and count incrementing (i.e. non-zero)
- The last sub-packet will have end flag = '1' and count incremented from previous packet count value.
- The original packet can be recreated by appending the payloads of the sub-packets and recalculating the length field.

4.2.2 Footer

The 4-byte footer is a checksum that may be used to verify the packet integrity. The checksum takes the form of a standard CRC-32 (32-bit cyclic redundancy check) and is calculated across the total packet (header and payload).

The details of the CRC are as follows:

- Truncated Polynomial (in hexadecimal): 0x04C1 1DB7
- Initial value: 0xFFFF FFFF
- Reflect input data: Yes
- Output CRC value reflected: Yes
- Final XOR value: 0xFFFF FFFF
- The CRC-32 algorithm is the same as used in zlib (<https://www.zlib.net/>), PKZip and ethernet.

Example CRC-32 calculation results:

Input word 0x0000 0000 yields 0x2144 DF1C

Input word 0x0102 0304 yields 0xB63C FBCD

4.3 Data Packets

The following packets are defined, with unique identifiers (Packet IDs).

Table 4-2: Packet Types

Packet ID	Packet Name	Description
0x00	SESSION START	Start of a session
0x01	SESSION END	End of a session
0x02	SCENE START	Start of an image scene
0x03	1. Only for HyperScape Imagers. Exposure Start	Time stamp at the instant when the exposure starts
0x04	LINE DATA	Pixel values for a single image line
0x05	THUMBNAIL LINE DATA	Pixel values for a single, subsampled, image line
0x06	<i>Reserved</i>	<i>Reserved</i>
0x07	TIME SYNCHRONISATION	Time stamped platform time.
0x08	TIME SYNCHRONISATION PPS	Time stamped at detection of PPS
0x09	<i>Reserved</i>	<i>Reserved</i>
0x0C to 0x7F	<i>Reserved</i>	<i>Reserved</i>
0x80	USER ANCILLARY DATA	User Ancillary Data
0x81 to 0x9F	<i>Reserved</i>	<i>Reserved</i>
0xA0 to 0xBF	IMAGER ANCILLARY DATA	Imager Ancillary Data
0xC0 to 0xFF	<i>Reserved</i>	<i>Reserved</i>

4.3.1 Session Start

The SESSION START packet is always the first packet written at the start of a session. It marks the start of the session, contains the Session ID and defines high-level mission details such as Platform and Instrument IDs.

Session Start		
Packet ID	0x00	
Flags	0x1	
Payload Length (bytes)	12	
Payload Field	Type	Description
Data Format	uint8	Major Version of the image data format
	uint8	Minor Version of the image data format
Platform ID	uint16	Platform (satellite) identifier Value is programmed by the user to identify the satellite
Instrument ID	uint16	Instrument (Imager) identifier Value is programmed by the user to identify the Imager
<i>Reserved</i>	uint16	<i>Reserved</i>

Session ID	uint32	Session identifier Automatically generated by the Imager to identify each session, in the form of a non-volatile 32-bit counter, incremented per session
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4.3.2 Session End

The SESSION END packet is always the last packet written and marks the end of the session.

Session End		
Packet ID	0x01	
Flags	0x1	
Payload Length (bytes)	4	
Payload Field	Type	Description
Session ID	uint32	Session identifier of session to end

4.3.3 Scene Start

The SCENE START packet always precedes any LINE DATA packets and marks the start of a scene within the session and provides the scene details.

Scene Start		
Packet ID	0x02	
Flags	0x1	
Payload Length (bytes)	12	
Payload Field	Type	Description
Scene Number	uint16	Starts from zero per session, auto increments
Scene Type	uint8	The type of image data to expect in this Scene: 0 : Snapshot 1 : Line Scan 2 : Snapshot Test Pattern 3 : Line Scan Test Pattern 4 : Video 5 : Line Scan High-Accuracy Mode ⁽¹⁾
<i>Reserved</i>	<i>uint8</i>	<i>Reserved</i>
Scene Height	uint24	Number of lines, with Binning factor applied. This value is calculated by dividing the sensor's row count by the Binning Factor value from the Imager Configuration packet (refer to section 4.3.10.2).
<i>Reserved</i>	<i>uint8</i>	<i>Reserved</i>
Scene Width	uint16	Pixels per line, with Binning factor applied. This value is calculated by dividing the sensor's column count by the Binning Factor value from the Imager Configuration packet (refer to section 4.3.10.2).
<i>Reserved</i>	<i>uint16</i>	<i>Reserved</i>

1. Only for HyperScape Imagers.

4.3.4 Exposure Start

The

1. Only for HyperScape Imagers.

Exposure Start packet always occurs before the Line Data packets and it is time stamped to store the instant at which the sensor exposure (integration) starts. There will be one

1. Only for HyperScape Imagers.

Exposure Start packet for each snapshot scene. Line scan scenes will have an Exposure Start Packet for each line.

Exposure Start		
Packet ID	0x03	
Flags	0x1	
Payload Length (bytes)	8	
Payload Field	Type	Description
Imager Time	uint64	Microsecond runtime counter value

4.3.5 Line Data

The LINE DATA packet contains the actual pixel data for the image line. In the case of line scan imagers, the pixel data is that of a specific band. For snapshot imagers, the pixel data is that of a single row in the snapshot image. The Line Length value is the number of sensor pixels per row divided by the Binning Factor value from the IMAGER CONFIGURATION packet (refer to section 4.3.10.2).

Line Data		
Packet ID	0x04	
Flags	0x1	
Payload Length (bytes)	varies	
Payload Field	Type	Description
Spectral Band	uint8	Spectral band number. Counts from zero (0). Value is always 0 for snapshot scenes
<i>Reserved</i>	<i>uint24</i>	<i>Reserved</i>
Line Number	uint24	Line number (within the range of Scene Height) Each band's line number autoincrements from 0.
<i>Reserved</i>	<i>uint8</i>	<i>Reserved</i>

Line Length	uint16	Number of pixels in this Line Data packet. This value is equal to the Scene Width.
Format	uint8	0 : Single Band, or Monochrome Snapshot 1 : RGB Bayer Snapshot (RG order) 2 : RGB Bayer Snapshot (GR order) 3 : RGB Bayer Snapshot (GB order) 4 : RGB Bayer Snapshot (BG order) Always 0 for Line Scan Imagers
Encoding	uint8	Bits per pixel and byte/word encoding 0 : 8-bit pixels per byte 1 : 10-bit pixels (packed, 4 pixels every 5 bytes) i.e., byte sequence [0x1a, 0x2b, 0x3c, 0x4d, 0x5e] represents four 10-bit pixels [0x31a, 0x30a, 0x0d3, 0x179] 2 : 10-bit pixels per 16-bit word 3 : 12-bit pixels (packed, 2 pixels every 3 bytes) i.e., byte sequence [0x1a, 0x2b, 0x3c] represents two 12-bit pixels [0xb1a, 0x3c2] 4 : 12-bit pixels per 16-bit word 5 : 16-bit pixels per 16-bit word
Pixel Data	varies	Pixel Data

4.3.6 Thumbnail Line Data

THUMBNAIL LINE DATA packets are generated when thumbnailing is enabled (refer to section 3.4.2.3). The format of the THUMBNAIL LINE DATA packet is the same as the LINE DATA packet in 4.3.5 except for the Line Length value which is the Scene Width divided by the Thumbnail Factor value from the IMAGER CONFIGURATION packet (refer to section 4.3.10.2).

Thumbnail Line Data		
Packet ID	0x05	
Flags	0x1	
Payload Length (bytes)	varies	
Payload Field	Type	Description
Spectral Band	uint8	Spectral band number. Counts from zero (0). Value is always 0 for snapshot scenes
<i>Reserved</i>	<i>uint24</i>	<i>Reserved</i>
Line Number	uint24	Line number (within the range of Scene Height) Each band's line number autoincrements from 0.
<i>Reserved</i>	<i>uint8</i>	<i>Reserved</i>
Line Length	uint16	This value is calculated by dividing the sensor's column count by the Thumbnail Factor value from the Imager Configuration packet (refer to section 4.3.10.2)

Format	uint8	0 : Single Band, or Monochrome Snapshot 1 : RGB Bayer Snapshot (RG order) 2 : RGB Bayer Snapshot (GR order) 3 : RGB Bayer Snapshot (GB order) 4 : RGB Bayer Snapshot (BG order) Always 0 for Line Scan Imagers
Encoding	uint8	Bits per pixel and byte/word encoding Value is always 0, thumbnails are always generated in 8-bit pixels.
Pixel Data	varies	Pixel Data

4.3.7 Time Synchronisation

The TIME SYNCHRONISATION packet allows the platform time to be stored with the Imager time (refer to section 3.4.1.6).

Time Sync		
Packet ID	0x07	
Flags	0x1	
Payload Length (bytes)	20	
Payload Field	Type	Description
Imager Time	uint64	Microsecond runtime counter value
Time Format ¹	uint8	Platform Time Format 0 : TAI 1 : UTC
<i>Reserved</i>	<i>uint24</i>	<i>Reserved</i>
Platform Time	uint64	Platform Time Value. Format varies.

1. More formats will be added in future

4.3.8 Time Synchronisation PPS

The TIME SYNCHRONISATION PPS packets are generated when the platform PPS signal is asserted. It allows the PPS signal to be time stamped in the image data stream. Refer to section 3.4.3.3. The user may have to delay triggering of the image capture until the PPS signal has been asserted once if a time reference is required at the start of the image data.

Time Sync PPS		
Packet ID	0x08	
Flags	0x1	
Payload Length (bytes)	8	
Payload Field	Type	Description
Imager Time	uint64	Microsecond runtime counter value

4.3.9 User Ancillary Data

USER ANCILLARY DATA packets allow the user to store data into the image data stream in real time during an active session. The platform transfers data to the Imager via the Control Interface and then this data is stored unaltered as the payload in USER ANCILLARY DATA packets. The maximum length of the User Data Payload field is 1024 bytes. Note that User Data ID's 0x80 to 0xFF are reserved.

User Ancillary Data		
Packet ID	0x80	
Flags	0x1	
Payload Length (bytes)	varies	
Payload Field	Type	Description
User Data ID	uint8	User specified ID to identify the ancillary data. 0x00 to 0x7F – User Specified ID 0x80 to 0xFF – Reserved
<i>Reserved</i>	<i>uint8</i>	<i>Reserved</i>
User Data Length	uint16	Length of the User Data field in bytes (1 to 1024)
User Data	varies	Maximum of 1024 bytes.

The contents (payload) of the USER ANCILLARY DATA packets is defined by the user although it is recommended that at least the following is considered:

- Platform attitude quaternions including the platform time within this data
- Platform ephemeris position and velocities including the platform time within this data

4.3.10 Imager Ancillary Data

The Imager automatically generates the following ancillary packets.

4.3.10.1 Imager Information

The IMAGER INFORMATION packet is inserted automatically at the start of a session.

Imager Information		
Packet ID	0xA0	
Flags	0x1	
Payload Length (bytes)	12	
Payload Field	Type	Description
Imager Product ID	uint24	Imager Product Identification Number
<i>Reserved</i>	<i>uint8</i>	<i>Reserved</i>
Imager Serial Number	uint16	Imager Serial Number
Firmware Major Version	uint8	Firmware Major Version
Firmware Minor Version	uint8	Firmware Minor Version
Software Major Version	uint8	Software Major Version
Software Minor Version	uint8	Software Minor Version

Baseline Number	uint16	Baseline Number for the Firmware and Software
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4.3.10.2 Imager Configuration

The IMAGER CONFIGURATION packet is inserted automatically once the Imager is configured, but before an image is taken. This packet is unique to the image data type: line scan, snapshot, or video.

For line scan image data:

Imager Configuration		
Packet ID	0xA1	
Flags	0x1	
Payload Length (bytes)	varies	
Payload Field	Type	Description
Line Period	uint32	Line Period (us) (inverse of the line rate) Integration time (per TDI stage) is derived from this value
Spectral Bands	uint8	Total number of spectral bands (set to n)
Exposure Time	uint32	Integration Time (μ s)
Band Setup	uint8 x n ¹	A list of bytes that specifies each band's TDI stages 0 : The band is disabled 1 : Line scan imaging without TDI 2 to 32 : Number of TDI stages
Binning Factor	uint8	Binning factor 0 : On-the-fly binning is disabled 2 : 2x2 pixel binning applied 4 : 4x4 pixel binning applied
Thumbnail Factor	uint8	Thumbnail reduction factor 0 : On-the-fly thumbnail generation disabled 2 : 2x reduction factor in each axis 4 : 4x reduction factor in each axis 8 : 8x reduction factor in each axis
Band Start Row	uint16 x n ¹	A list of 16-bit words that specify the location on the sensor (row) to be used for this band's effective line.
Band Centre Wavelength	uint16 x n ¹	A list of 16-bit words that specify the configured centre wavelength of each band in nanometre.
Scan Direction	uint8	0 – Positive 1 – Negative
Reserved	varies	Reserved bytes (padding to a 4-byte boundary)

1. n is equal to the "Spectral Bands" payload field

For snapshot image or video data:

Imager Configuration		
Packet ID	0xA4	
Flags	0x1	
Payload Length (bytes)	12	
Payload Field	Type	Description
Frame Interval	uint32	Inter-frame time (μ s) for burst snapshot or video captures.
Exposure Time	uint32	Integration Time (μ s)
Binning Factor	uint8	Binning factor 0 : On-the-fly binning is disabled 2 : 2x2 pixel binning applied 4 : 4x4 pixel binning applied
Thumbnail Factor	uint8	Thumbnail reduction factor 0 : On-the-fly thumbnail generation disabled 2 : 2x reduction factor in each axis 4 : 4x reduction factor in each axis 8 : 8x reduction factor in each axis
Reserved	uint16	Reserved

4.3.10.3 Sensor Configuration

The SENSOR CONFIGURATION packet is inserted automatically once the Imager is configured, but before an image is taken.

The packet format is dependent on the Imager. Please refer to the Imager's User Manual for the definition of the Sensor Configuration packet.

4.3.10.4 Imager Telemetry

The IMAGER TELEMETRY packet is inserted every 500 ms during an active session, while the sensor is enabled.

Imager Telemetry		
Packet ID	0xA3	
Flags	0x1	
Payload Length (bytes)	12	
Payload Field	Type	Description
Imager Time	uint64	Microsecond runtime counter value
Sensor Temperature	int8	Temperature in °C
Reserved	uint24	Reserved

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